and increase of $O_2$(Table 13.5). This condition changes the carboxylase role of RUBISCO into oxygenase. $C_2$ Cycle takes place in chloroplast, peroxisome and mitochondria. RUBP is converted into PGA and a 2C-compound phosphoglycolate by Rubisco enzyme in chloroplast. Since the first product is a 2C-compound, this cycle is known as **$C_2$ Cycle**. Phosphoglycolate by loss of phosphate becomes glycolate. Glycolate formed in chloroplast enters into peroxisome to form glyoxylate and hydrogen peroxide. Glyoxylate is converted into glycine and transferred into mitochondria. In mitochondria, two molecules of glycine combine to form serine. Serine enters into peroxisome to form hydroxy pyruvate. Hydroxy pyruvate with help of NADH + $H^+$ becomes glyceric acid. Glyceric acid is cycled back to chloroplast util ising ATP and becomes Phosphoglyceric acid (PGA) and enters into the Calvin cycle (PCR cycle). Photorespiration does not yield any free energy in the form of ATP. Under certain conditions 50% of the photosynthetic potential is lost because of Photorespiration (Figure 13.20).

### 13.15.1 Significance of photorespiration

1. Glycine and Serine synthesised during this process are precursors of many biomolecules like chlorophyll, proteins, nucleotides.
2. It consumes excess NADH + $H^+$ generated.
3. Glycolate protects cells from Photo oxidation.

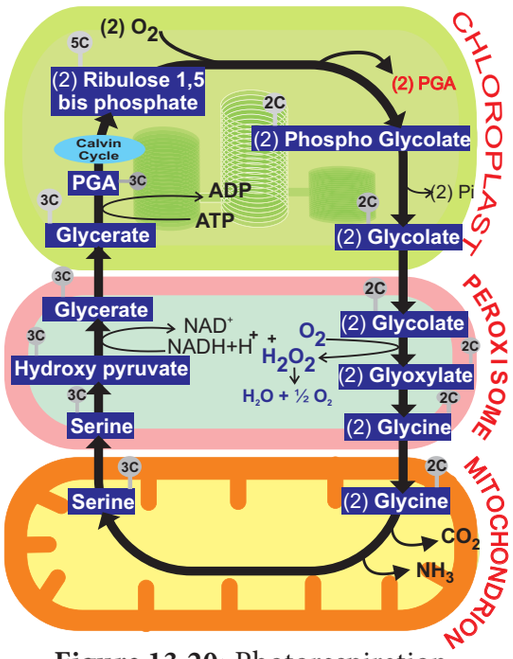



**Figure 13.20:** Photorespiration

### 13.15.2 Carbon Dioxide Compensation Point

When the rate of photosynthesis equals the rate of respiration, there is no exchange of oxygen and carbon dioxide and this is called as carbon dioxide **compensation point**. This will happen at particular light intensity when exchange of gases becomes zero. When light is not a limiting factor and atmospheric $CO_2$ concentration is between 50 to 100 ppm the net exchange is zero.

## 13.15 Factors affecting Photosynthesis

In 1860, **Sachs** gave three cardinal points theory explaining minimum, optimum and maximum factors that control photosynthesis.

**Table 13.5: Differences between Photorespiration and Dark Respiration**

| Photorespiration | Dark respiration |
|---|---|
| 1. It takes place in photosynthetic green cells | 1. It takes place in all living cells |
| 2. It takes place only in the presence of light | 2. It takes place all the time |
| 3. It involves chloroplast, peroxisome and mitochondria | 3. It involves only mitochondria |
| 4. It does not involve Glycolysis, Kreb's Cycle, and ETS | 4. It involves glycolysis, Kreb's Cycle and ETS |
| 5. Substrate is glycolic acid | 5. Substrate is carbohydrates, protein or fats |
| 6. It is not essential for survival | 6. Essential for survival |
| 7. No phosphorylation and yield of ATP | 7. Phosphorylation produces ATP energy |
| 8. $NADH_2$ is oxidised to $NAD^+$ | 8. $NAD^+$ is reduced to $NADH_2$ |
| 9. Hydrogen peroxide is produced | 9. Hydrogen peroxide is not produced |
| 10. End products are $CO_2$ and PGA | 10. End products are $CO_2$ and water |



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